Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 Feb 2026
Team ID	
Project Name	Strategic Product Placement & Sales Analysis (FMCG Sector)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

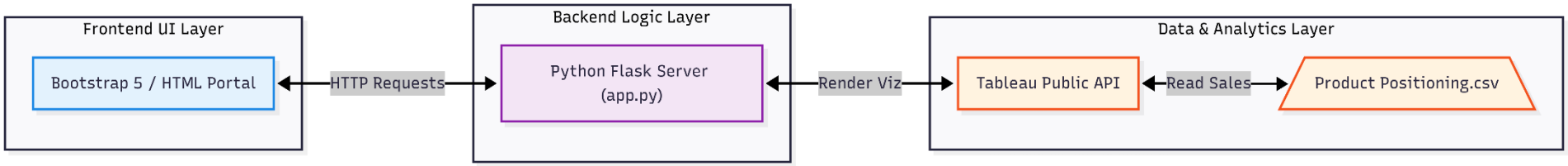

Table 1: Components & Technologies

This table breaks down every piece of software you used to build the portal.

S.No	Component	Description	Technology
1	User Interface	Responsive Web UI providing the analytics portal dashboard and data story view.	HTML5, CSS3, Bootstrap 5, JavaScript
2	Application Logic-1	Backend web server routing and request handling.	Python (Flask Framework)
3	Application Logic-2	Data ingestion, cleaning, and statistical correlation calculations.	Python (Pandas, NumPy)
4	Application Logic-3	Visual analytics engine rendering dynamic charts and filters.	Tableau Desktop / Public
5	Database	Flat-file data storage containing historical sales and placement records.	CSV (Product Positioning.csv)
6	Cloud Database	Cloud hosting for the interactive visualizations and packaged workbooks.	Tableau Public Server
7	File Storage	Local storage for HTML templates (index.html) and static report assets.	Local Filesystem

S.No	Component	Description	Technology
8	External API-1	API used to securely embed and render the visualizations inside the web portal.	Tableau JavaScript API (viz_v1.js)
9	Infrastructure	Application deployed for local development and mentor demonstration.	Localhost Server (Werkzeug/Flask)

Table 2: Application Characteristics

This table explains the non-functional technical design of your system.

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Core frameworks used for backend routing and frontend UI styling.	Flask (Python), Bootstrap 5
2	Security Implementations	Application runs on a local environment for data privacy. Tableau embeds use secure HTTPS transmission.	Localhost Isolation, HTTPS / TLS
3	Scalable Architecture	3-tier architecture separating the UI (Bootstrap), Logic (Flask), and Analytics (Tableau).	Python, JS, Tableau

S.No	Characteristics	Description	Technology
4	Availability	The analytics engine is highly available via cloud hosting, while the portal runs locally on-demand.	Tableau Public Cloud
5	Performance	Dashboards use client-side rendering (iFrames) to reduce backend server load and ensure fast response times.	HTML iFrame, JavaScript

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d